MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 8056

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Triode, miniature, metal shell, medium Mu

Outline --- 4-4 (EIA) - See figure 1

Base --- E5-65

Envelope --- MT-4

Cathode --- Coated unipotential

Base connections:

Pin No.	---	1	2	3	4	5	6	7	8	9	10	12
Element	---	int con	a	int con	g	int con	int con	int con	k	int con	h	h

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ef	Eb	Ec	Ehk	Rk	Rk	Ik	Ic	Pp	TE	Alt
Unit:	V	Vdc	Vdc	v	Ohms	Meg	mAdc	mAdc	W	°C	ft
Maximum:	6.9	50	+2,-55	100	---	10	15	2.0	0.45	150	100,000
Minimum:	5.7	---	---	---	---	(see note 1)	---	---	---	---	---
TEST CONDITIONS:	6.3	24	---	---	100	---	---	---	---	---	---

GENERAL:

Qualification - Required

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T14:49Z
Check the source to verify that this is the current version before use.

METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS		UNIT
						MIN	MAX	
	Qualification inspection							
1002	Barometric pressure, reduced	Pressure = 8.0 ±0.5 mmHg; voltage = 250 Vac	6.5	Code E	---	---	---	---
1041	Shock (1)	No voltages; peak acceleration 500 G; time duration = 0.65 ±0.15 milliseconds	---	---	---	---	---	---
1031	Sweep-frequency vibration-fatigue	Eb = Ec = 0; Rk = 0	6.5	Code E	---	---	---	---
---	Shock (1) and sweep-frequency vibration-fatigue test end points:							
1306	Transconductance (1)		---	---	ΔSmt	---	10	%
1266	Total grid current		---	---	Ic	---	-0.1	µAdc
1366	Heater-cathode leakage		---	---	Ihk	---	5	µAdc
1031	Sweep-frequency vibration (1)	3 to 6 kHz 6 to 15 kHz	--- ---	--- ---	Ep Ep	--- ---	75 750	mv mv
	Quality conformance inspection, part 1							
1301	Heater current		0.4	II	If	125	145	mA
1336	Heater-cathode leakage		0.4	II	Ihk	---	3	µAdc
1266	Total grid current	Eb = 40 Vdc; Rg = 1.0 Meg; Ec = -2.0 Vdc; Rk = 0 (see note 2)	0.4	II	Ic	0	-0.05	µAdc
1256	Electrode current (1) (anode)		0.4	II	Ib	6.7	10.7	mAdc
1256	Electrode current (2) (anode)	Ec = -10.0 Vdc; Rk = 0	0.4	II	Ib	---	50	µAdc
1306	Transconductance (1)		0.4	II	Sm	6,500	8,500	µmhos
1201	Short and discontinuity detection	See note 3	0.4	II	---	---	---	---
	Quality conformance inspection, part 2							
1211	Insulation of electrodes	E(g to all) = -100 Vdc E(a to all) = -100 Vdc	2.5 2.5	I I	R R	1,000 1,000	--- ---	Meg Meg
1306	Transconductance (2)	Ef = 5.7 V	1.0	I	Sm Ef,	5,700	---	µmhos
1316	Amplification factor		6.5	Code E	Mu	8	14	---
1306	Transconductance (3)	Ef = 5.7 V	4.0	Code F	ΔSm Ef	---	15	%

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T14:49Z
Check the source to verify that this is the current version before use.

METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 2 -Continued							
1256	Electrode current (3) (anode)	Eb = 12.0 Vdc; Rg = 0.033 Meg; Rk = 0; Cg = 0.5 µF	6.5	Code E	Ib	4.2	7.4	mAdc
1306	Transconductance (4)	Eb = 12.0 Vdc; Rg = 0.033 Meg; Rk = 0; Cg = 0.5 µF	6.5	Code E	Sm	6,600	9,400	µmhos
1331	Direct-interelectrode capacitance	No shield	6.5	Code E	Cin	3.4	4.6	pF
					Cout	1.4	2.0	pF
					Chk	1.1	1.7	pF
					Cg-p	1.8	2.4	pF
					Cp-k	0.26	0.42	pF
1031	Sweep-frequency vibration (1)	Rp = 2,000 ohms; 4 G (see note 4)						
		3 to 6 kHz	6.5	Code E	Ep	---	50	mv
		6 to 15 kHz	6.5	Code E	Ep	---	500	mv
1031	Sweep-frequency vibration (2)	Rp = 2,000 ohms; 4 G; F = 0.05 to 3 kHz	6.5	See note 5	Ep	---	50	mVac
1041	Shock (2)	1,000 G; Eb = 50 Vdc; Ec = -3.6 Vdc; Ehk = +100 V; Rk = 0 (see note 6)	---	See note 5	---	---	---	---
---	Shock (2) end points:							
1306	Transconductance (1)		---	---	ΔSm t	---	10	%
1266	Total grid current		---	---	Ic	---	-0.1	µAdc
1336	Heater-cathode leakage		---	---	Ihk	---	5	µAdc
1031	Sweep-frequency vibration (1)	3 to 6 kHz	---	---	Ep	---	75	mv
		6 to 15 kHz	---	---	Ep	---	750	mv

METHOD	REQUIREMENT OR TEST	CONDITIONS	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 3					
1506	Heater-cycling life	Ef = 8.5 V; Ehk = -100 Vdc; Eb = Ec = 0; Rk = 0; 1 min "on" 2 min "off"	---	---	---	---
---	Heater-cycling life-test end point:					
1336	Heater-cathode leakage		Ihk	---	15	µAdc
1516	Stability life	Eb = 50 Vdc; Ec = -3.6 Vdc; Ehk = +100 Vdc; Rg = 10 Meg; Rk = 0; TA = room	---	---	---	---
---	Stability life-test end point:					
1306	Transconductance (1), change of individual tubes		ΔSm t	---	10	%

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T14:49Z
Check the source to verify that this is the current version before use.

METHOD	REQUIREMENT OR TEST	CONDITIONS	SYMBOL	LIMITS		UNIT
				MIN	MAX	
	Quality conformance inspection, part 3 -Continued					
1501	Intermittent life	Eb = 50 Vdc; Ec = -3.6 Vdc; Ehk = +100 Vdc; Rg = 10 Meg; Rk = 0; TA = room TE = +150°C (min) (see notes 8 and 9)	---	---	---	---
---	Intermittent life-test end points: (500 hours)	See note 7				
---	Inoperatives		---	---	---	---
1266	Total grid current		Ic	---	-0.1	µAdc
1301	Heater current		If	125	147	mA
1306	Transconductance (1), change of individual tubes		ΔSm t	---	15	%
1336	Heater-cathode leakage		Ihk	---	5	µAdc
1211	Insulation of electrodes	E = -100 Vdc	Rg-all Rp-all	500 500	--- ---	Meg Meg
1306	Transconductance (1), average change		Avg ΔSm t	---	10	%
---	Total defectives		---	---	---	---
1511	Cathode interface life	Ef = 6.9 V (see notes 7 and 9)	R1	---	25	Ohms

NOTES:

1. Maximum grid-current resistance applies for either fixed-bias or cathode-bias operation at metal -shell temperatures up to 150°C.
2. This test shall be performed at the conclusion of the holding period.
3. The TUT shall be tapped three times in each of two planes 90° to 128° apart, using a hand tapper consisting of a bakelite rod 1/8-inch (3.18 mm) in diameter and 7-inches (177.80 mm) in length with a rubber tip 1-inch (25.4 mm) long. The rubber tip consists of gum tubing 1/8-inch (3.18 mm) ID and 3/32-inch (2.38 mm) wall thickness with an average Durometer rating of 35 ±5 Short A, or equal.
4. The TUT shall be vibrated in the plane perpendicular to its axis (X plane) through a frequency range of 3,000 to 15,000 Hz. Sweep time shall be approximately 7.0 seconds, and the rate of change of frequency shall be approximately linear. The TUT shall be rotated to find the direction of vibration in the X plane which gives the highest output reading. The holding fixture shall be in accordance with Drawing 184-JAN.
5. This test shall be conducted on the initial lot and thereafter on a lot approximately every 12 months. When one lot has passed, the 12-month rule shall apply. In the event of lot failure, the lot shall be rejected and the succeeding lots shall be subjected to this test until a lot passes. MIL-STD-105, sample size code letter E, shall apply.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T14:49Z
Check the source to verify that this is the current version before use.

NOTES: -Continued

6. A grid resistor of 10 megohms shall be added; however, this resistor shall not be used when a thyratron-type short indicator is used. The holding fixtures shall be in accordance with Drawing 184-JAN.

7. The allowable defectives per characteristic shall not be greater than 1 for the first sample and shall not be greater than 3 for combined samples.

8. Envelope temperature (TE) requirements, when measured in accordance with the temperature by conduction-band measurement (method 1226), or thermocouple elements welded to shell, will be satisfied if a TUT having bogey Ib (+5 percent) under normal test conditions, is determined to operate at or above minimum specified temperature at any position in the life-test rack.

9. The life-test sample shall consist of 20 tubes, and not more than 1 tube failure shall be permitted. In the event of rejection of the first sample due to failure of more than 1 tube, a second sample of 40 tubes shall be selected from the lot. Acceptance shall be based on the combined first and second samples. The total tube failures from the combined first and second samples shall not exceed three.

10. Revision letters are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Air Force - 99
DLA - ES

User activities:
Navy - AS, MC, CG, OS
Air Force - 11

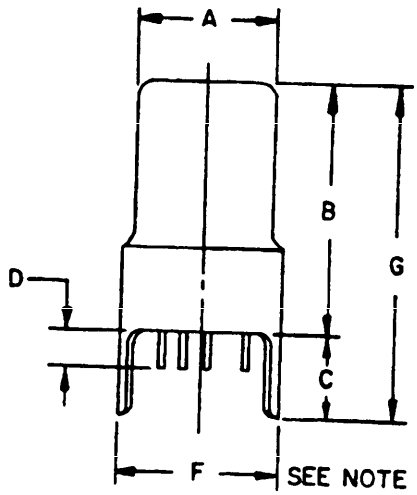
Preparing activity:
Navy - EC

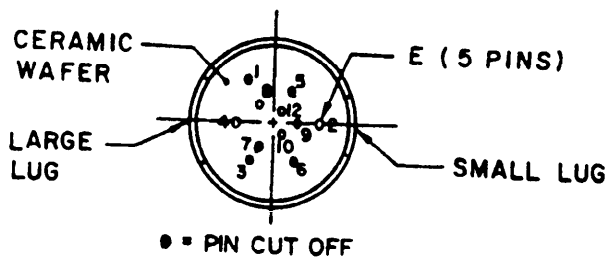
Agent:
DLA - ES

(Project 5960-3235)

Source: https://assist.dla.mil -- Downloaded: 202610-15T14:49Z
Check the source to verify that this is the current version before use.

Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 1		
A		0.400(10.16)
B		0.625(15.88)
C	0.175(4.45)	0.210 (5.33)
D	0.100(2.54)	0.130 (3.30)
E	0.015(0.38)dia	0.017(0.43)dia
Quality conformance inspection, part 2		
F		0.435(10.25)
G		0.800(20.32)





NOTE: Maximum OD of .440(11.18 mm) is permitted along the .190(4.83 mm) lug length.

FIGURE 1. Outline drawing of electron tube type 8056.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T14:49Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE NAVY

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

NO POSTAGE
NECESSARY
IF MAILED
IN THE
UNITED STATES

BUSINESS REPLY MAIL

FIRST CLASS PERMIT NO. 12503 WASHINGTON D. C

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE NAVY

Commander
Naval Electronic Systems Command
Defense Standardization Program Branch
Department Of The Navy
Washington, D.C. 20360

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T14:49Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE
MIL-E-1/1490E

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

S/N 0102-014-1802

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T14:49Z
Check the source to verify that this is the current version before use.